

CS 472 Lab 2 Report - Michael Podolsky

Phase 1 Report Requirements

- Screenshot showing the test_coverage_lab folder and a successful pytest run on my own machine

The screenshot shows a code editor with a file explorer on the left and a terminal at the bottom. The file explorer shows the test_coverage_lab folder with files like .coverage, .gitignore, LICENSE, pytest.ini, README.md, requirements-dev.txt, requirements.txt, .pytest_cache, instance, test.db, models, __init__.py, account.py, __pycache__, tests, __init__.py, test_account.py, __pycache__, fixtures, and account_data.json. The terminal shows the command `python3 -m pytest --cov=models --cov-report=term-missing` and the output of the test run, including a coverage report.

```
test_account.py
73 def test_invalid_role_assignment():
74     with pytest.raises(DataValidationError):
75
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78 # =====
79 # Test: Password Hashing
80 # Author: Michael Podolsky
81 # Date: 2026-09-09
82 # Description: Ensure passwords are properly hashed and that password verification works correctly.
83 # =====
84
85 def test_password_hashing():
86     """Test that passwords are hashed and verified correctly"""
87     account = Account(name="Michael Podolsky", email="podolm1@unlv.nevada.edu")
88     password = "secret-password"
89
90     account.set_password(password)
91
92     # The password is stored as a hash, not as plaintext
93     assert account.password_hash is not None
94     assert account.password_hash != password
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96     # Verification succeeds for the correct password only
97     assert account.check_password(password) is True
98     assert account.check_password("wrong-password") is False
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```

First, the password is stored as a hash rather than plaintext: **password_hash** is populated and does not equal the original password. Second, **check_password()** correctly distinguishes passwords, returning **True** for the password that was set and **False** for an incorrect one.

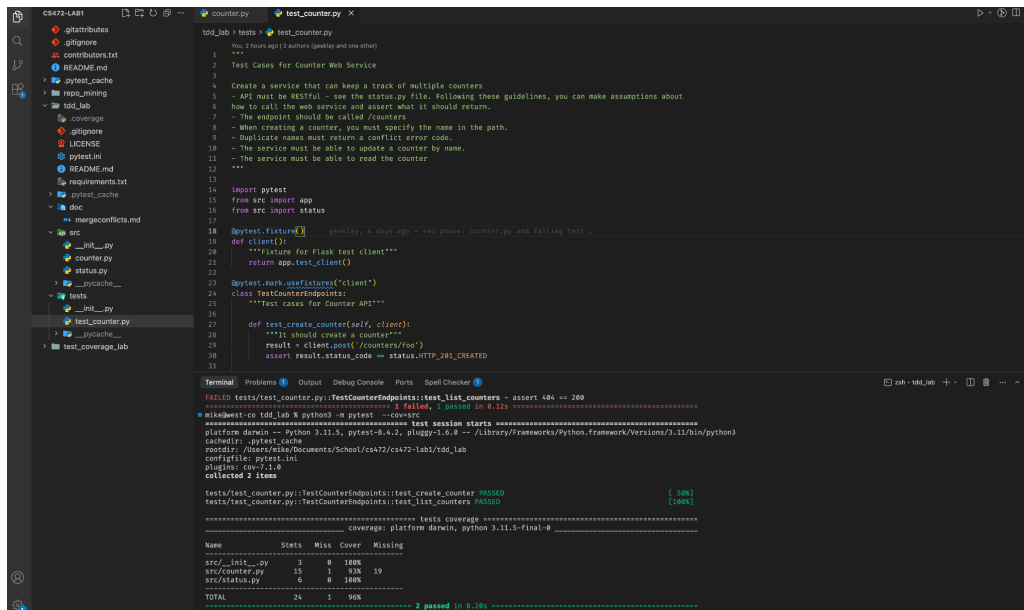
- Link to the teammate's PR I reviewed and to my review comments:
<https://github.com/kenner1-unlv/Group-2/pull/77>
<https://github.com/kenner1-unlv/Group-2/pull/77#pullrequestreview-5239503863>
<https://github.com/kenner1-unlv/Group-2/pull/77#pullrequestreview-5249858807>
- A short summary of the feedback I received and how I addressed it:

I received a review from Glen, who stated that all the changes looked appropriate and met the assignment requirements.

- Link to the approved PR or merge commit:
<https://github.com/kenner1-unlv/Group-2/commit/df24d9628227ca93053d40e8f142b8e30759b7e2>

Phase 2 Report Requirements

- Screenshot showing the `tdd_lab` folder and pytest running on my own machine:



The screenshot shows a VS Code editor with a project structure on the left. The 'tdd_lab' folder is expanded, showing files like `__init__.py`, `counter.py`, `status.py`, and `test_counter.py`. The `test_counter.py` file is open in the editor, showing test cases for the Counter Web Service. The terminal window at the bottom shows the output of running `pytest` on the tests, indicating that all tests passed and coverage is 100%.

```
FAILED tests/test_counter.py::TestCounterEndpoints::test_list_counters - assert 106 == 200
===== 1 failed, 1 passed, 10 8.25s =====
platform darwin -- Python 3.11.5, pytest-8.4.2, pluggy-1.6.0 -- /Library/Frameworks/Python.framework/Versions/3.11/bin/python3
cachedir: .pytest_cache
rootdir: /Users/nike/Documents/school/cs472-lab1/tdd_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 2 items

tests/test_counter.py::TestCounterEndpoints::test_create_counter PASSED [ 50%]
tests/test_counter.py::TestCounterEndpoints::test_list_counters PASSED [100%]

===== tests coverage =====
coverage: platform darwin, python 3.11.5-final-0

Name          Stats    Miss Cover Missing
src/_init_.py    3      0 100%
src/counter.py  15      1 93%  19
src/status.py    6      0 100%
TOTAL           24      1 96%
```

- Setup PR link: <https://github.com/kenner1-unlv/Group-2/pull/47>
- Setup issue link: <https://github.com/kenner1-unlv/Group-2/issues/46>
- Link to my issue: <https://github.com/kenner1-unlv/Group-2/issues/49>
- Link to my PR: <https://github.com/kenner1-unlv/Group-2/pull/50>

- Brief explanation of feature being tested:

List all counters (**GET /counters**). The Counter API could create counters via **POST /counters/<name>**, but provided no way to read them back. Once created, a counter's value was inaccessible. This feature adds a **GET /counters** endpoint that returns every counter currently stored along with its value, responding with **HTTP_200_OK**.

- RED evidence:
<https://github.com/kenner1-unlv/Group-2/pull/50#issuecomment-5669608500>
- GREEN evidence:
<https://github.com/kenner1-unlv/Group-2/pull/50#issuecomment-5669612937>
- REFACTOR justification:

None necessary. The implementation is a single return with no branching and no logic to extract; **jsonify** serializes **COUNTERS** directly. It shares no duplicated logic with **create_counter()**. **counter_exists()** does not apply, since listing has no name to look up and no error path.

- A short summary of the feedback I received and how I addressed it:

I received a review from Glen, who stated that all the changes looked appropriate and met the assignment requirements.

- Link to teammate's PR I reviewed and my comments:
<https://github.com/kenner1-unlv/Group-2/pull/92>
<https://github.com/kenner1-unlv/Group-2/pull/92#pullrequestreview-5250180190>
<https://github.com/kenner1-unlv/Group-2/pull/92#pullrequestreview-5250704732>
- A brief statement of the evidence I examined while reviewing (for example: test results, RED/GREEN evidence, coverage report):

I reviewed the diff to **tdd_lab/src/counter.py** and **tdd_lab/tests/test_counter.py**, along with the RED evidence in the PR description showing **assert 500 == 404** from an unhandled **KeyError**, the GREEN run showing 8 tests passing, and the coverage report at 97% with line 33 still uncovered. I then applied the change to a clean copy of upstream/main and ran the suite locally, reproducing the same 8 passed and 97% coverage figures, which confirmed the reported results were accurate.

- Link to the approved PR or merge commit:
<https://github.com/kenner1-unlv/Group-2/commit/29aa70516aef131017223be20cfca4b8427cb8a2>

Phase 3 Report Requirements

- Link to my fork: <https://github.com/MikePodo/cs472-lab1>
- Evidence that my fork was synchronized with the upstream repository:

```
mike@west-co tdd_lab % git fetch upstream
remote: Enumerating objects: 51, done.
remote: Counting objects: 100% (51/51), done.
remote: Compressing objects: 100% (46/46), done.
remote: Total 51 (delta 21), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (51/51), 11.16 KiB | 141.00 KiB/s, done.
From https://github.com/kenner1-unlv/Group-2
  5ffdfb..a331fb1  main      -> upstream/main
* [new branch]      setup-ci-lab-kenner1-unlv -> upstream/setup-ci-lab-kenner1-unlv
mike@west-co tdd_lab % git merge upstream/main
Updating 5ffdfb..a331fb1
Fast-forward
 .github/workflows/ci.yml | 34 ++++++
 ci_lab/.gitignore        | 64 ++++++
 ci_lab/LICENSE           | 21 +++++
 ci_lab/README.md         | 91 ++++++
 ci_lab/pytest.ini        | 5 ++
 ci_lab/requirements-dev.txt | 4 ++
 ci_lab/requirements.txt  | 4 ++
 ci_lab/src/__init__.py   | 12 +++
 ci_lab/src/counter.py    | 131 ++++++
 ci_lab/src/status.py     | 8 +++
 ci_lab/tests/__init__.py | 0
 ci_lab/tests/test_counter.py | 292 ++++++
12 files changed, 666 insertions(+)
create mode 100644 .github/workflows/ci.yml
create mode 100644 ci_lab/.gitignore
create mode 100644 ci_lab/LICENSE
create mode 100644 ci_lab/README.md
create mode 100644 ci_lab/pytest.ini
create mode 100644 ci_lab/requirements-dev.txt
create mode 100644 ci_lab/requirements.txt
create mode 100644 ci_lab/src/__init__.py
create mode 100644 ci_lab/src/counter.py
create mode 100644 ci_lab/src/status.py
create mode 100644 ci_lab/tests/__init__.py
create mode 100644 ci_lab/tests/test_counter.py
mike@west-co tdd_lab %
```

- Evidence that the ci_lab starter workflow ran successfully for you:

```
cachedir: .pytest_cache
rootdir: /Users/mike/Documents/School/cs472/cs472-lab1/ci_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 22 items

tests/test_counter.py::TestCounterEndpoints::test_create_counter PASSED [ 4%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_duplicate_counter PASSED [ 9%]
tests/test_counter.py::TestCounterEndpoints::test_retrieve_existing_counter PASSED [ 13%]
tests/test_counter.py::TestCounterEndpoints::test_return_404_for_non_existent_counter PASSED [ 18%]
tests/test_counter.py::TestCounterEndpoints::test_increment_counter PASSED [ 22%]
tests/test_counter.py::TestCounterEndpoints::test_prevent Updating non-existent counter PASSED [ 27%]
tests/test_counter.py::TestCounterEndpoints::test_delete_counter PASSED [ 31%]
tests/test_counter.py::TestCounterEndpoints::test_prevent deleting non-existent counter PASSED [ 36%]
tests/test_counter.py::TestCounterEndpoints::test_reset_all_counters PASSED [ 40%]
tests/test_counter.py::TestCounterEndpoints::test_list_all_counters PASSED [ 45%]
tests/test_counter.py::TestCounterEndpoints::test_handle_invalid http methods PASSED [ 50%]
tests/test_counter.py::TestCounterEndpoints::test_get_total_counters PASSED [ 54%]
tests/test_counter.py::TestCounterEndpoints::test_top_n_counters PASSED [ 59%]
tests/test_counter.py::TestCounterEndpoints::test_bottom_n_counters PASSED [ 63%]
tests/test_counter.py::TestCounterEndpoints::test_set_counter_to_value PASSED [ 68%]
tests/test_counter.py::TestCounterEndpoints::test_prevent negative counter values PASSED [ 72%]
tests/test_counter.py::TestCounterEndpoints::test_reset_single_counter PASSED [ 77%]
tests/test_counter.py::TestCounterEndpoints::test_prevent resetting non-existent counter PASSED [ 81%]
tests/test_counter.py::TestCounterEndpoints::test_get_total_number_of_counters PASSED [ 86%]
tests/test_counter.py::TestCounterEndpoints::test_counters_greater_than threshold PASSED [ 90%]
tests/test_counter.py::TestCounterEndpoints::test_counters_less_than threshold PASSED [ 95%]
tests/test_counter.py::TestCounterEndpoints::test_validate_counter_name PASSED [100%]

===== tests coverage =====
coverage: platform darwin, python 3.11.5-final-0

Name           Stmts  Miss  Cover   Missing
-----
src/__init__.py    3      0  100%
src/counter.py   87      5   94%   71, 85, 96, 99-100
src/status.py      7      0  100%
TOTAL             97      5   95%

===== 22 passed in 0.19s =====
```

- Link to my issue: <https://github.com/kenner1-unlv/Group-2/issues/54>
- Link to my PR: <https://github.com/kenner1-unlv/Group-2/pull/55>
- Link to the workflow excerpt:
<https://github.com/MikePodo/cs472-lab1/blob/e92c94b9226556c7396bfa8dbe261d0dbd5d5d96/.github/workflows/ci.yml#L10-L16>

- Link to the failing CI log and my root-cause summary:
<https://github.com/kenner1-unlv/Group-2/actions/runs/34909742000/job/104194352681?pr=55>

This failed because on Python 3.9, **zip()** accepts no keyword arguments at all, so the call raises `TypeError`.

- Link to the fix commit or PR diff:
<https://github.com/kenner1-unlv/Group-2/pull/55/changes/4c4a0d4df5dfd919b4ec6b83deaa386635933e9a>
- Link to successful CI run: <https://github.com/kenner1-unlv/Group-2/pull/55/checks>
- Link to teammate's PR I reviewed and commented on:
<https://github.com/kenner1-unlv/Group-2/pull/95>
<https://github.com/kenner1-unlv/Group-2/pull/95#pullrequestreview-5251441576>
- A brief statement of the evidence I examined while reviewing (for example: CI logs, test results, workflow diff):

I reviewed the net workflow diff adding the **status** job to `.github/workflows/ci.yml`, then parsed the merged file to confirm the job graph, checking that it declares **needs: [lint, test, coverage]**, an if: **always() && github.event_name == 'pull_request'** condition, and **permissions: pull-requests: write**. I checked the linked failing workflow run and the bot comments posted on the PR as evidence the job works on success.

Reflection

This lab taught me that a test passing doesn't mean it's actually testing anything. I wrote one that would have passed even if the code did nothing, and I only caught it because I wrote the test before the code. The CI part was the most eye-opening, since my tests passed on my laptop but failed on Python 3.9 in the pipeline, which I never would have caught without the matrix.